

Thermally Assisted Nanoscratching of Monocrystalline Silicon up to
800 °C: Tribology and Subsurface Transformations

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Abstract:

The expanding applications of micro-electro-mechanical systems (MEMS) and the stringent surface quality requirements for high-precision X-ray optical components have brought unprecedented attention to the machining challenges of monocrystalline silicon. The conventional monocrystalline silicon machine is under scrutiny, as monocrystalline silicon is difficult to machine without damage at room temperature. Thermal softening at elevated temperatures may enhance surface quality, but the

1 underlying mechanisms remain unclear. Therefore, this study explores monocrystalline
2 silicon's tribological behavior, material removal mechanisms and subsurface defect
3 evolution during ramp nanoscratching from ambient to 800 °C. Three distinct wear
4 regimes were identified: below 400 °C, scratching caused microcracking and chipping.
5 At 600 °C, material removal shifted to ductile ploughing without fracture. At 800 °C,
6 lateral force increased by 2.5 times, with material bulging, pile-ups and intense shear
7 bands, indicating excessive plastic slip. Subsurface analysis showed a deep defect zone
8 of nanocrystalline and amorphous silicon beneath a SiO₂ layer. In contrast, 600 °C
9 produced the thinnest defect layer, suggesting an optimal thermal window. Molecular
10 dynamics simulations confirmed temperature-dependent phase evolution during
11 scratching. These findings identify a favorable temperature range for damage-
12 minimized nanomachining of monocrystalline silicon.

13 **Keywords:** Monocrystalline silicon, Ramp-nanoscratching, Thermal-mechanical
14 coupling, Subsurface damage

15

1. Introduction

Monocrystalline silicon (Si) is widely used in semiconductor manufacturing, energy conversion systems and optical applications due to its excellent mechanical, electronic and optical properties[1–3]. However, during nanoscale mechanical processing, Si undergoes phase transformations, plastic deformation and defect formation, leading to subsurface damage such as amorphization, dislocations, stacking faults and microcracks. These defects degrade mechanical strength, electrical performance and optical quality, making damage-free machining a critical challenge[4,5]. Multi-energy-field-assisted techniques such as chemical mechanical polishing (CMP) and laser-assisted processing (LAP) have been developed to improve Si machinability[6–8]. These methods leverage thermal and mechanical effects to modify Si's deformation behavior. However, the fundamental mechanisms governing temperature-dependent phase transformations, plastic deformation, and subsurface damage evolution remain unclear. Understanding these mechanisms is essential for optimizing material removal efficiency, minimizing machining-induced defects and improving surface quality in precision Si processing.

At room temperature (RT, 25°C), monocrystalline silicon exhibits a highly load-dependent mechanical response[9,10], where different processing methods induce distinct deformation behaviors. On a macroscopic scale, Si is characterized by high hardness and brittleness[11]. However, at the micro and nanoscale, Si undergoes various phase transformation pathways, often accompanied by high-pressure phase

transformations (HPPT), which are considered the primary mechanism governing its plastic deformation[12]. Under different loading conditions, Si follows distinct phase transformation sequences. During nanoindentation, the typical transformation pathway is Si-I (diamond cubic) $\rightarrow$ Si-II (β -tin, high-pressure phase) during loading and Si-II $\rightarrow$ Si-III (body-centered cubic) / Si-XII (rhombohedral) / amorphous Si (a-Si) during unloading[13]. In contrast, during nanoscratching, the dominant transformation is Si-I $\rightarrow$ a-Si, and at higher loads, a-Si may further transform into other metastable phases depending on the stress state and strain rate[14]. In addition to phase transformations, dislocations, stacking faults and microcracks are frequently observed[15], leading to severe subsurface damage. These complex damage and transformation behaviors pose significant challenges for damage-free machining of Si. Meanwhile, extensive molecular dynamics simulations at room temperature have provided insights into the plastic deformation mechanisms of Si. Most studies suggest that HPPT dominates plasticity, while dislocation motion and twinning also contribute to the overall deformation process[16–20]. These findings highlight the intricate interplay between phase transformations, dislocation activities and stress-induced amorphization, which collectively dictate the mechanical response of Si under different loading conditions.

At elevated temperatures, monocrystalline silicon gradually transitions from a brittle material to a plastic material. Korte et al. observed that under the same loading conditions, microcracks appear at low temperatures, whereas at 500°C, deformation is primarily governed by twinning and partial dislocations, with no microcrack

1 formation[21]. Additionally, studies have shown that as temperature increases, both
2 hardness and Young's modulus of Si decrease, indicating enhanced plasticity and
3 improved machinability[22]. Given that Si is increasingly used in high-temperature
4 applications, extensive research has been conducted in recent years to investigate its
5 mechanical behavior across different temperature ranges. Studies have identified a
6 brittle-to-ductile transition (BDT) threshold in Si, typically between 300–400°C[23,24].
7 Above this threshold, hardness decreases exponentially and crack formation is
8 significantly suppressed. Unlike at room temperature, where HPPT dominates plasticity,
9 Kiran et al. reported that at 200°C, twinning becomes the primary plastic deformation
10 mechanism[25]. Similarly, Eremenko and Nikitenko found that at 350°C, plastic
11 deformation is mainly governed by dislocation motion and microcrack formation[26].
12 Schaffar et al. conducted nanoindentation experiments from RT to 950°C and confirmed
13 trends observed in previous studies[27]. However, they further investigated the
14 influence of oxidation on the mechanical response of Si, revealing that oxidation at high
15 temperatures reduces dislocation activity, increasing hardness. Molecular dynamics
16 simulations by Zhang et al. demonstrated that at high temperatures, dislocations
17 nucleate in the subsurface, promoting the formation of metastable phases and providing
18 deeper insights into the mechanical response of Si at elevated temperatures[28]. In situ
19 transmission electron microscopy (TEM) studies by Li et al. [29]revealed that at 600°C,
20 stress-induced amorphous Si undergoes recrystallization, growing along the crystalline
21 orientation and inducing dislocation formation. Additionally, Sun et al.[30] investigated
22 laser-assisted defect healing in Si, identifying a critical energy threshold: below this

1 threshold, thermal energy can repair defects while exceeding it introduces additional
2 damage. These findings highlight the complex interplay between temperature, phase
3 transformations, dislocation activity and oxidation, which collectively govern the high-
4 temperature deformation behavior of Si.

5 Previous studies have suggested that high-temperature conditions may facilitate
6 damage-free ultra-precision machining of monocrystalline silicon. However, the
7 damage mechanisms of Si under high-temperature scratching conditions remain unclear.
8 To address this gap, this study employs both experimental investigations and molecular
9 dynamics simulations to systematically analyze the damage mechanisms of Si during
10 scratching at different temperatures. In this work, we conduct linear ramp scratching
11 experiments on Si along the [110] crystallographic direction[31], which is more prone
12 to defect formation. A Berkovich diamond indenter with a tip radius of approximately
13 500 nm is used to apply a gradually increasing load from 0 to 80 mN. The mechanical
14 response, surface deformation and subsurface plastic deformation of Si are
15 systematically analyzed. Furthermore, molecular dynamics simulations are performed
16 to interpret the experimental observations theoretically. This study identifies the
17 temperature-dependent transitions in deformation mechanisms and subsurface damage
18 evolution during silicon nanoscratching, determining an optimal temperature range for
19 achieving balanced plasticity and minimal subsurface damage.

2. Materials and methods

2.1 High-temperature ramp nanoscratching

Ramp nanoscratch tests were conducted on monocrystalline silicon using a multifunctional tribometer (Rtec/MFT-5000) with a custom-built high-temperature chamber. The silicon wafer exhibited a surface roughness of 0.3 nm, providing an atomically flat surface. The high-resolution transmission electron microscopy (HRTEM) image (Fig. 1(a)) and fast Fourier transform (FFT) pattern (Fig. 1(b)) confirm the defect-free diamond cubic structure of the as-received sample. Testing employed a Berkovich diamond indenter (tip radius ~ 500 nm, presented in Fig. 1(f)) with linearly increasing normal force (0-80 mN) at a constant scratch velocity of 25 $\mu\text{m/s}$ over 100 μm length. The indenter was oriented with the leading edge (edge of the triangular pyramid) positioned in the scratching direction, as shown in Fig. 6(a). It should be noted that previous studies have reported that material damage mechanisms during machining are influenced by tool orientation angles[32–34]; however, since this is not the focus of our current investigation, all experiments were conducted with identical indenter orientation to control variables and isolate temperature effects. An initial 2 mN threshold force ensured proper indenter-surface contact. All scratches were aligned along the [110] crystallographic orientation (Fig. 1(c)), which exhibits higher susceptibility to defect formation. The high-temperature chamber was equipped with thermoelectric heating elements and integrated with a recirculating water cooling system to achieve stable temperature control (Fig. 1(d, e)), with an accuracy of $\pm 0.1^\circ\text{C}$.

Temperature was monitored using calibrated S-type thermocouples, ensuring uniformity across the sample. Prior to each experiment, the chamber was heated at a rate of 20°C/min to the target temperature, after which the indenter was introduced and the system was allowed to equilibrate for 20 minutes to ensure thermal stability. Experiments were conducted at five temperatures: 25°C (room temperature), 200°C, 400°C, 600°C, and 800°C. Following each test, the chamber was cooled at a controlled rate of 20°C/min until reaching 25°C to minimize thermal shock and preserve sample integrity. Due to equipment limitations, all high-temperature nanoscratch tests were conducted in an ambient air atmosphere. To provide context for the temperature-dependent mechanical response of monocrystalline silicon during scratching, Table 1 summarizes literature-reported approximate values of hardness (H) and Young's modulus (E) at the tested temperatures. These data were derived from nanoindentation studies[27] that investigated two (100)-oriented monocrystalline silicon wafers using maximum loads up to 50 mN (comparable to those in this work). The results demonstrate a clear thermal softening effect, H decreases significantly above ~400°C. This aligns with the BDT and informs the expected deformation behavior in our experiments. Normal force (F_z), tangential force (F_x) and their ratio (F_x/F_z) were continuously recorded (Fig. 1(g-i)). The F_x/F_z ratio serves as a critical indicator of deformation resistance and reveals transitions between different deformation mechanisms across temperature regimes.

Table 1: Approximate temperature-dependent hardness and Young's modulus of monocrystalline silicon from literature[27].

Temperature (°C)	Hardness (GPa)	Young's Modulus (GPa)
25	12.4	173
200	11.5	164
400	10.2	152
600	6.0	128
800	2.3	110

2.2 Surface and subsurface morphology characterization

Following the nanoscratch tests, the surface morphology of the scratched regions was analyzed using a Bruker Contour GT-K white light interferometer, which provided three-dimensional surface profiles to assess the topographical changes induced by scratching. Additionally, a scanning electron microscope (SEM, Merlin, ZEISS) was employed to examine the detailed surface features, including material pile-up, fracture patterns and wear characteristics.

Cross-sectional TEM specimens were prepared using a Helios 600i focused ion beam (FIB) system. Specimens were extracted perpendicular to scratches at 50 μm from the starting point. To minimize ion-beam damage and ensure uniform thickness, a multi-stage milling protocol was implemented: initial thinning at 30 kV (beam current reducing from 0.79 nA to 80 pA) with $\pm 1^\circ$ specimen tilt to reduce curtaining effects until reaching ~ 100 nm thickness, followed by final polishing at 5 kV (15 pA) with $\pm 3^\circ$ tilt for 3 minutes per angle.

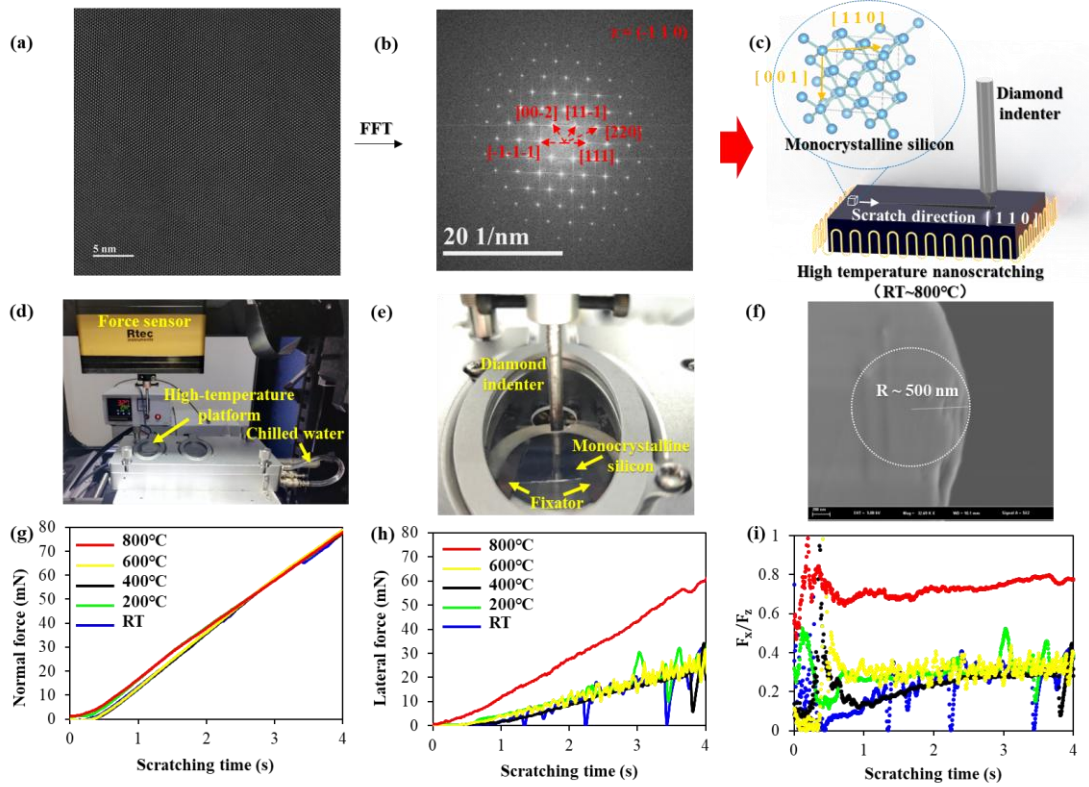


Figure 1. (a) HRTEM image of the atomic arrangement of original monocrystalline silicon. (b) FFT pattern confirming the diamond cubic phase along the [110] orientation. (c) Schematic of the nanoscratch test setup with a Berkovich indenter. (d) High-temperature nanoscratch apparatus. (e) Experimental setup showing indenter-specimen configuration. (f) SEM image of Berkovich tip ($R \approx 500$ nm). (g) Normal force vs. time at different temperatures. (h) Tangential force vs. time at different temperatures. (i) F_x/F_z vs. time at different temperatures.

Before TEM observation, the specimens were cleaned using a modified Fischione 1020 plasma cleaner to eliminate residual contamination and ensure optimal imaging conditions[35]. The prepared TEM specimens were examined using a FEI Titan Cubed

Themis G2 transmission electron microscope, which provided high-resolution imaging and diffraction analysis. The subsurface amorphization, dislocation evolution and damage depth were systematically analyzed to elucidate the deformation mechanisms under different temperature conditions.

2.3 Molecular dynamics simulation

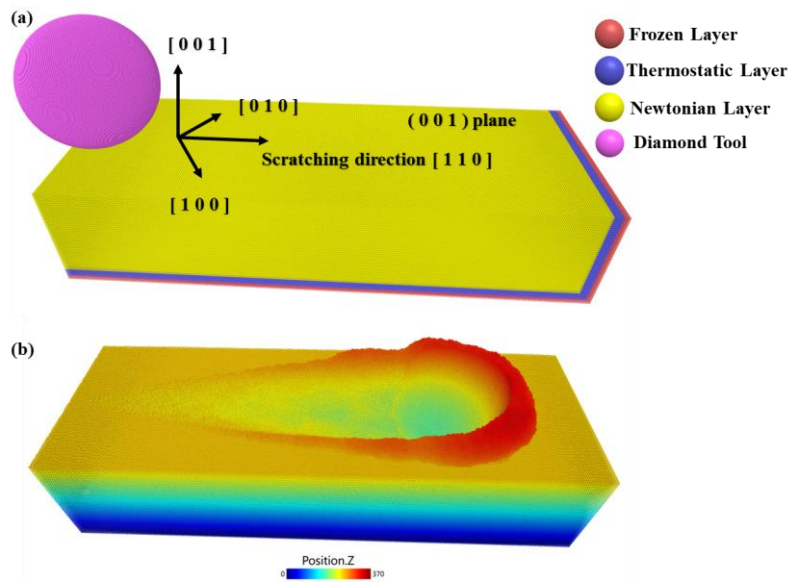
To further investigate the high-temperature deformation mechanisms of monocrystalline silicon, molecular dynamics simulations were performed using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS)[36]. The simulation model is illustrated in Fig. 2 (a), where a monocrystalline silicon sample with dimensions of $40.26 \times 115.58 \times 24.72 \text{ nm}^3$ was constructed, consisting of 5,793,375 atoms. The simulations were conducted at four different temperatures: 25°C, 600°C, 800°C and 1000°C.

The simulation temperatures were selected to focus on those temperatures at which distinct mechanistic changes were likely to occur according to experimental observations. Therefore, 200 and 400°C were deliberately omitted from the molecular dynamics simulation to conserve computational resources. While experimental tests were performed up to 800°C, an additional simulation at 1000°C was included to extrapolate trends and understand extreme temperature behavior, particularly the thermally-activated plasticity mechanisms and potential recrystallization phenomena beyond the experimental range.

1 The silicon model was constrained by a 1 nm-thick frozen layer at both the bottom
2 and rear end, with an adjacent 1 nm-thick thermostatic layer for temperature regulation.
3 The remaining volume was designated as the Newtonian layer. Periodic boundary
4 conditions were applied along the x-direction, with non-periodic shrink-wrapped
5 conditions in the y- and z-directions. Consistent with the experimental setup, the
6 nanoscratch simulations were performed on silicon's (001) surface along the [110]
7 crystallographic direction. Although a Berkovich indenter was used in experiments, a
8 diamond spherical indenter with a radius of 14 nm was employed as the scratching tool
9 in simulations to avoid computational complexities associated with sharp edges while
10 maintaining the essential contact mechanics. This geometrical simplification is justified
11 by previous studies[37]. The spherical geometry also eliminates the orientation-
12 dependent effects inherent to pyramidal indenters, allowing us to isolate the
13 temperature-dependent deformation mechanisms more effectively. Given the
14 significantly higher hardness of diamond compared to silicon, the indenter was modeled
15 as a rigid body, neglecting its deformation.

16 It should be noted that molecular dynamics simulations are limited by
17 computational power and truncation errors, making it impossible to replicate the exact
18 scale used in experiments. To address these limitations, we employed a relatively large-
19 scale model to minimize boundary effects and better approximate experimental
20 conditions[38–40]. Previous studies have shown that simulations under relatively high
21 speeds can still reveal the main mechanisms of nanometric scratching because silicon

1 exhibits low sensitivity to loading rate according to previous studies[41–43]. Our
 2 simulation primarily focuses on the influence of temperature on material plasticity
 3 mechanisms, thus minimizing the impact of size and loading rate discrepancies.
 4 Therefore, a relatively high scratching velocity was adopted to enhance computational
 5 efficiency while maintaining accuracy. The indenter was assigned two velocity
 6 components: a horizontal velocity of 100 m/s along the scratching direction and a
 7 vertical velocity of -10 m/s. The total scratch length was 71 nm, with a maximum
 8 penetration depth of 7.1 nm. The Erhart-Albe potential was employed to model Si-Si
 9 atomic interactions[37,44], while Si-C interactions between silicon and the diamond
 10 indenter were described using the Morse potential. The simulation protocol included:
 11 (1) energy minimization to eliminate residual stress, (2) thermal equilibration at the
 12 designated temperature and (3) scratching with the indenter initially positioned 5 Å
 13 above the surface. Temperature was maintained using the temp/rescale method, with a
 14 2.5 fs time step balancing computational efficiency and accuracy.



1 Figure 2. (a) Molecular dynamics simulation model of nanoscratching on
2 monocrystalline silicon, showing the (001) surface and the scratching direction along
3 [110]. (b) Side view of the silicon sample after scratching.

4
5 Post-processing and data analysis were conducted using Python, MATLAB and
6 OVITO[45], enabling visualization and quantitative evaluation of the deformation
7 mechanisms, including subsurface defect evolution, phase transformation and material
8 removal characteristics.

9
10

3. Results and discussion

3.1 Surface morphology in high-temperature scratching

The surface morphology of the scratched regions at different temperatures was captured using a Bruker Contour GT-K white light interferometer, as shown in Fig. 3(a)–(e). The corresponding cross-sectional profiles at the scratch endpoints, extracted perpendicular to the scratching direction, are also presented to provide a more detailed view of the material deformation. A short secondary scratch perpendicular to the primary scratch direction can be observed at the scratch endpoint in all cases. This phenomenon occurs due to the elastic recovery of the silicon sample upon unloading, which causes the indenter to rebound slightly when it is lifted without active force control.

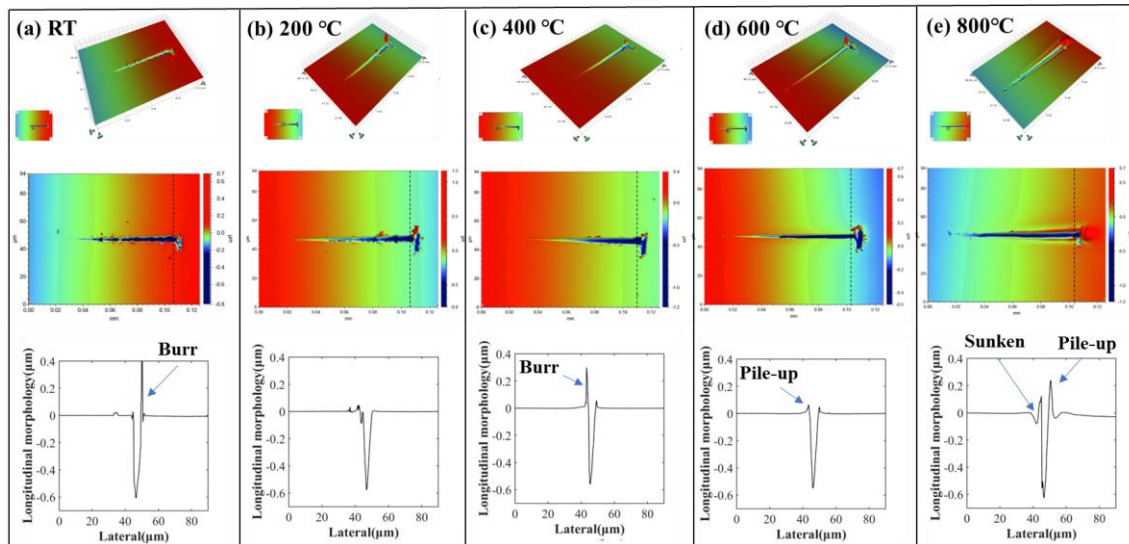


Figure 3. Surface morphology and cross-sectional profiles of scratched regions at

different temperatures. (a) RT (25°C), (b) 200°C, (c) 400°C, (d) 600°C and (e) 800°C.

The top row presents 3D surface profiles obtained using a white light interferometer.

The middle row shows corresponding 2D top views. The bottom row displays cross-sectional profiles near the scratch endpoints.

At RT and 200°C (Fig. 3(a) & (b)), scratch edges exhibit irregular burr-like features characteristic of brittle fracture. Rapid crack propagation causes sudden energy release, removing material beyond the primary scratch path and deteriorating surface quality. Visible cracks within the scratch track indicate that applied loads exceed the plastic deformation threshold of silicon at these temperatures.

With increasing temperature, brittle fracture and edge chipping progressively diminish. At 400°C (Fig. 3(c)), edges become smoother with only minor burrs. At 600°C (Fig. 3(d)), edges transition to smooth profiles with notable material pile-up along both sides, providing direct evidence of a shift from fracture-dominated to plasticity-dominated deformation. At 800°C (Fig. 3(e)), the pile-up effect becomes even more pronounced and an additional sunken region appears adjacent to the pile-up. This suggests that plastic flow is further enhanced at higher temperatures. Moreover, a significant accumulation of displaced material can be observed at the scratch endpoint, forming a semi-circular bulge. This material accumulation phenomenon, observed only at 800°C, reflects significantly altered flow characteristics compared to lower temperature conditions. This observation further supports the F_x/F_z analysis, which

will be discussed in the following sections.

Table 2: Temperature-dependent material removal amount.

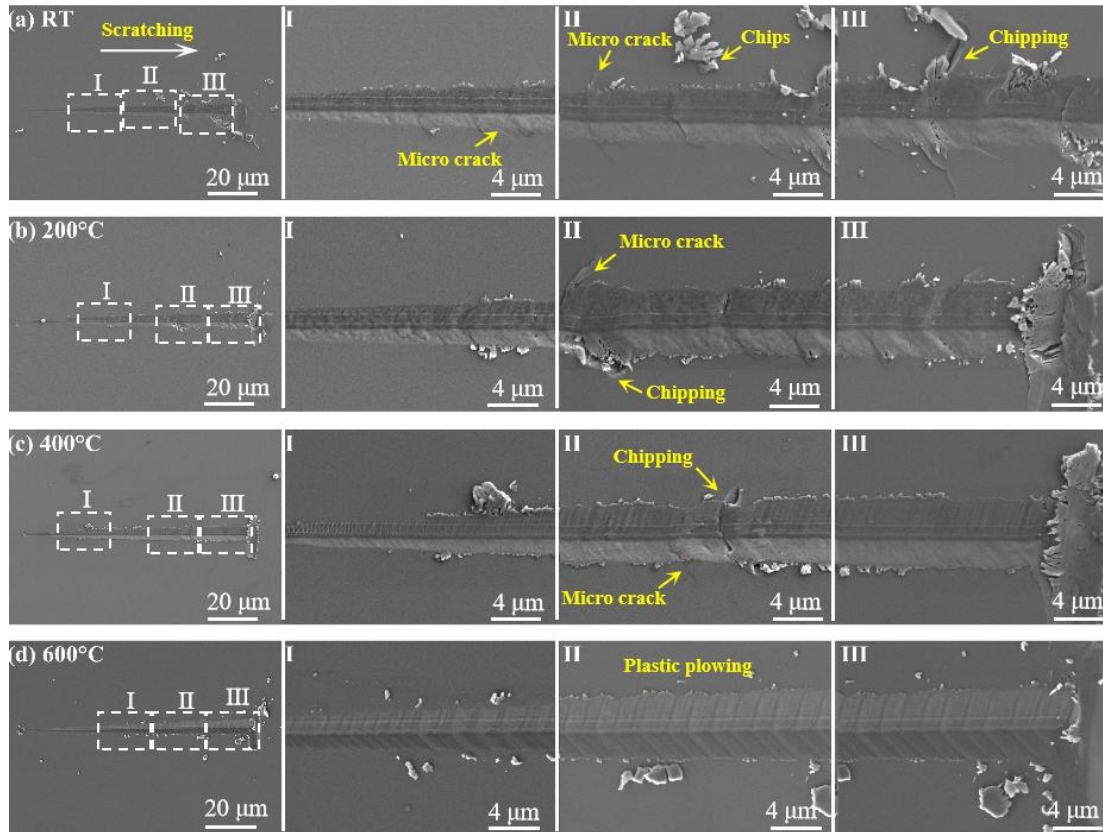
Temperature (°C)	Average material removal (μm^3)	Standard deviation (μm^3)
25	69.854	± 5.37
200	73.188	± 3.067
400	63.06	± 5.438
600	51.271	± 8.401
800	83.998	± 9.656

Table 2 summarizes the quantitative analysis of material removal volumes as a function of temperature, calculated by integrating the cross-sectional area below the original surface along the scratch length. At room temperature (25°C), the material removal volume reaches $69.854 \pm 5.37 \mu m^3$, primarily attributed to brittle fracture-dominated wear mechanisms that generate significant chip formation and lateral cracking. As the temperature increases to 200°C and 400°C, the removal volumes are $73.188 \pm 3.067 \mu m^3$ and $63.06 \pm 5.438 \mu m^3$, respectively, indicating a transitional regime where brittle and ductile mechanisms coexist. At 600°C, the material removal volume decreases to its minimum value of $51.271 \pm 8.401 \mu m^3$, corresponding to the optimal ductile regime where plastic flow without fracture becomes dominant, resulting in more efficient material removal with reduced surface damage. Notably, at 800°C, the removal volume increases dramatically to $83.998 \pm 9.656 \mu m^3$ —a 64% increase compared to 600°C. This substantial increase is primarily attributed to severe thermal

softening (hardness reduction from ~ 6 GPa at 600°C to 2.3 GPa at 800°C , as shown in Table 1), which allows the indenter to penetrate deeper into the substrate. The excessive material softening at 800°C promotes pronounced plastic flow and pile-up rather than efficient material removal. It should be noted that the scratch depth is also influenced by the pressure-induced phase transformation from crystalline silicon (Si-I) to amorphous silicon (a-Si), which exhibits a different atomic packing density ($\sim 20\%$ volume change). However, the relative contribution of this phase transformation effect versus purely mechanical penetration cannot be definitively separated from these macroscopic measurements alone. The atomic-scale mechanisms governing phase transformation and their temperature dependence will be elucidated through molecular dynamics simulations in Section 3.3, where the interplay between thermally activated plasticity and pressure-induced amorphization can be directly visualized and quantified.

SEM analysis (Fig. 4(a-d)) revealed significant temperature-dependent evolution of deformation mechanisms. At RT and 200°C , scratch edges exhibit extensive microcracks, edge breakage and detached chips, confirming brittle fracture-dominated material removal. Internal cracks within the scratch region are further evidence of brittle deformation. This fracture behavior corresponds directly to the large fluctuations in tangential force (Fig. 1(e)), resulting from intermittent crack propagation and rapid stress relaxation. At 400°C , microcracks and edge breakage diminish significantly, indicating a transition toward ductile deformation despite residual minor fractures. At 600°C , a complete transformation occurs—scratch edges become entirely smooth

1 without visible microcracks or breakage, while tangential force curves show smooth
2 continuous profiles without fluctuations. Prominent pile-up formations along both sides
3 of the scratch track confirm plastic flow dominance. The absence of brittle features
4 demonstrates silicon's excellent plasticity and machinability at this temperature.



5
6 Figure 4. SEM images of scratched surfaces at different temperatures: (a) RT (25°C),
7 (b) 200°C, (c) 400°C and (d) 600°C. Insets I–III show magnified views of the scratch
8 edges.

9
10 At 800°C, the scratch morphology undergoes a fundamental transition (Fig. 5).
11 Chip formation is completely suppressed, replaced by pronounced pile-up and bulging

characteristic of enhanced plastic flow. Close examination reveals three key features:

- (1) continuous material displacement through plastic plowing rather than fracture;
- (2) significant material accumulation along scratch edges;
- (3) dense shear bands extending from scratch edges into surrounding material.

These shear bands (Fig. 5, inset III) represent a critical microstructural feature unique to high-temperature silicon deformation, forming along preferred crystallographic planes with $\sim 0.2 \mu\text{m}$ spacing between adjacent bands. This pattern indicates collective dislocation motion rather than the random dislocation activity observed at lower temperatures. Despite plastic deformation dominance, localized microcracks appear at the scratch endpoint, likely resulting from either excessive stress accumulation during final scratching stages or thermal stress induced by post-experiment cooling.

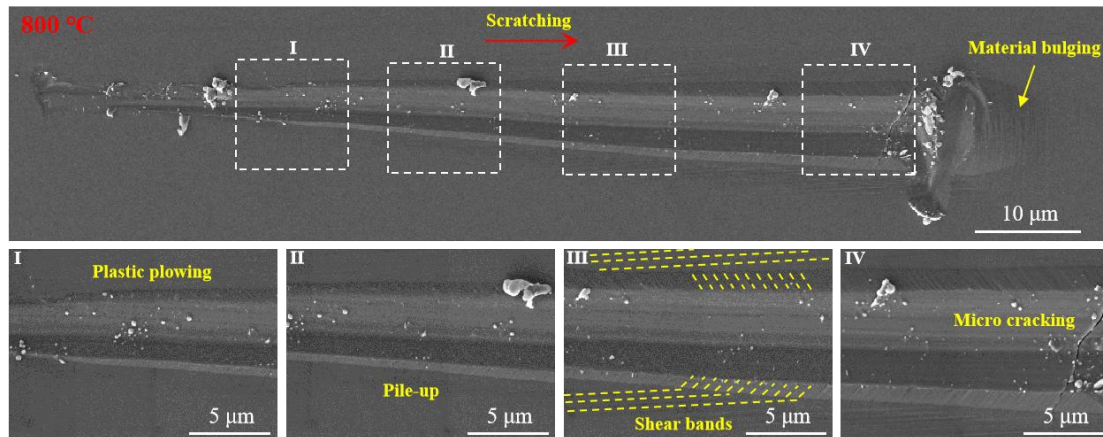


Figure 5. SEM images of the scratched surface at 800°C. The top image shows an overview of the scratch morphology. Insets I–IV provide magnified views of key deformation features.

With increasing scratch depth at 800°C, shear bands and extended deformation zones proliferate, confirming plastic slip as the dominant mechanism. This results in a significant increase in scratching resistance and F_x/F_z , which reaches approximately 2.5 times the value observed at RT–600°C. The increased F_x/F_z is attributed to the fact that material removal is suppressed and deformation occurs primarily via elastic-plastic flow rather than fracture. Thus, at 800°C, the nanoscale machining behavior and material removal mechanism of monocrystalline silicon undergoes a fundamental transition, marking the onset of severe plastic deformation and material softening.

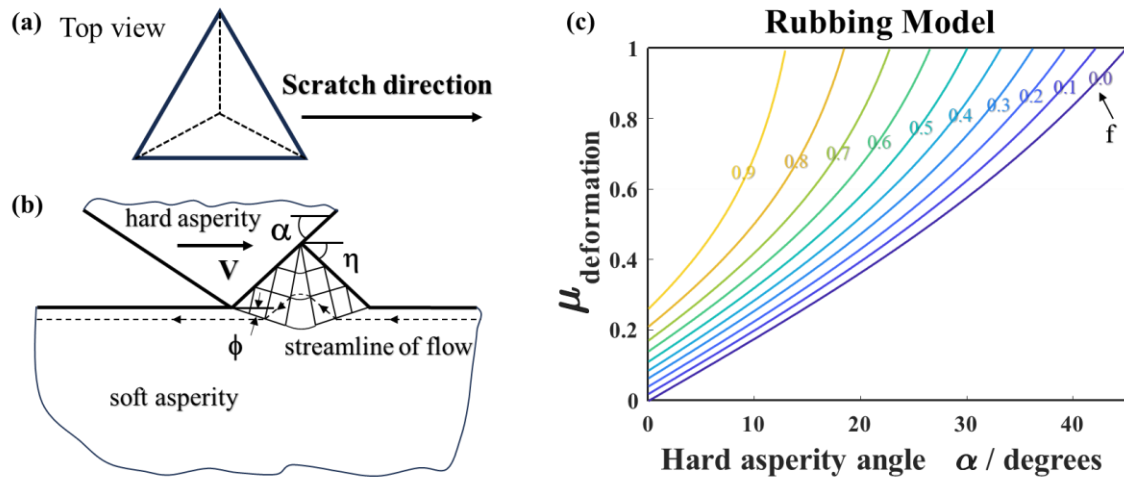


Figure 6. (a) Schematic representation of Berkovich indenter orientation; (b) Schematic diagram of Challen and Oxley rubbing model; (c) Theoretical friction coefficient $\mu_{\text{deformation}}$ as a function of hard asperity angle α for different interfacial film strengths f .

To elucidate the observed increase in the F_x/F_z ratio at 800°C, where F_x/F_z rises

to approximately 0.78 (2.5 times higher than at lower temperatures), we analyze the contributions from adhesion and deformation using established tribological models. This threshold behavior aligns with the temperature-dependent softening and deformation mode transitions in monocrystalline silicon.

The adhesion component can be understood through the Bowden-Tabor model[46], which describes the friction coefficient as:

$$\mu_{adhesion} = \frac{Q_{adhesion}}{P_{adhesion}} = \frac{\tau * A}{H * A} = \frac{\tau}{H}$$

with τ denoting the interfacial shear strength and H the hardness of the substrate. From RT to 600°C, both τ and H decrease proportionally due to thermal softening, as H is proportional to the material's yield stress and τ is typically related to the shear yield strength k (with k/H remaining fairly constant[46,47]). This constancy supports the stable $\mu_{adhesion}$ in silicon from RT to 600°C. At 800°C, however, the formation of a ~20 nm-thick amorphous SiO₂ layer (as confirmed by EELS in Fig. 9(c)) alters this balance. While SiO₂ and Si exhibit similar surface energies, leading to comparable adhesion tendencies[48,49], the underlying Si substrate softens dramatically ($H \approx 2.3$ GPa at 800°C [27], compared to ~10 GPa at 400°C). In contrast, SiO₂ remains more thermally stable, with its hardness decreasing linearly from 8.85 ± 0.05 GPa at RT to approximately 8 GPa at 800°C under comparable loads (200 mN)[50]. Consequently, the effective τ (influenced by the SiO₂ surface) decreases less than H (dominated by the Si bulk, given scratching depths $\gg 20$ nm), resulting in a higher $\mu_{adhesion}$ at 800°C. This elevated adhesion contributes to the observed increase in F_x/F_z , as the model's

1 $\mu_{adhesion}$ approximates the tangential-to-normal force ratio under adhesive conditions.

2 The deformation (ploughing) component further amplifies the F_x/F_z increase at
3 800°C, building on the adhesion effects to yield an overall ratio $\mu_{total} \approx \mu_{adhesion} +$
4 $\mu_{deformation}$. Silicon undergoes a BDT around 700-900°C (depending on loading rate)
5 [51,52], with fracture toughness increasing nearly fivefold due to enhanced dislocation-
6 mediated plasticity, as evidenced by our TEM and MD results. At such temperatures,
7 phase transformations are suppressed, and deformation is dominated by dislocations
8 rather than phase transition. To quantify the ploughing effect, we apply the Challen and
9 Oxley rubbing model [53] for rigid-plastic materials under sliding conditions, which
10 yields a friction coefficient $\mu_{deformation}$ analogous to the F_x/F_z ratio in ploughing-
11 dominated scratching.

12 Although silicon exhibits elastic-plastic behavior overall, our experimental
13 observations at 800°C, combined with literature evidence of enhanced ductility, justify
14 this approximation for qualitative analysis: TEM images reveal extensive dislocation
15 networks and shear bands (Fig. 9) indicative of steady-state plastic flow with minimal
16 elastic recovery; literature reports show that at the BDT, silicon's fracture toughness
17 increases nearly fivefold and enlarged plastic zones; and SEM profiles (Fig. 5) display
18 pronounced pile-up and material bulging without significant elastic rebound,
19 resembling the standing wave deformation assumed in the model. These features
20 collectively demonstrate that silicon behaves similarly to a rigid-plastic material under
21 high-temperature scratching conditions, validating the model's application.

The model considers a rigid asperity (indenter) sliding over a soft material with shear flow stress k , interfacial film strength f ($0 \leq f \leq 1$, ratio of film shear strength to k), and asperity angle α (here $\sim 24.5^\circ$ for the Berkovich indenter) as shown in Fig. 6 (b).

The friction coefficient due to deformation is given by:

$$\mu_{deformation} = \frac{A \sin \alpha + \cos(\cos^{-1} f - \alpha)}{A \cos \alpha + \sin(\cos^{-1} f - \alpha)}$$

where:

$$A = 1 + \frac{\pi}{2} + \cos^{-1} f - 2\alpha - 2 \sin^{-1} \left(\frac{\sin \alpha}{\sqrt{1-f}} \right)$$

SEM and TEM observations reveal a relatively smooth and dense oxide surface at 800°C , indicating a high f (e.g., $f \approx 0.5$, as the compact SiO_2 film provides superior shear strength compared to the softened Si's k). As shown in Fig. 6 (c), higher f and fixed α yield elevated $\mu_{deformation}$ (e.g., ~ 0.65 for $f=0.5$ vs. ~ 0.45 for $f=0.1$). Below 600°C , fracture and balanced chip removal offset ploughing, stabilizing F_x/F_z ; at 800°C , excessive softening suppresses removal, leading to pronounced pile-up, material bulging and shear bands, which dominate $\mu_{deformation}$ and thus elevate F_x/F_z .

This analysis highlights a critical temperature threshold ($\sim 600\text{-}800^\circ\text{C}$), closely linked to the BDT, where thermal softening, coupled with the formation of an oxide film, shifts deformation from efficient ductile removal to excessive plastic flow—even in the absence of oxide-induced increases in f , the BDT alone would elevate F_x/F_z at

1 800°C by promoting enhanced plasticity and material flow. Future studies could
2 incorporate finite element modeling to quantify elastic contributions, validate f values
3 under varying atmospheres, and explore the oxide film's influence—for instance,
4 experiments in inert gas environments may reveal a reduction in F_x/F_z (estimated at
5 40-50% based on model predictions with lower f) due to decreased f , as the absence of
6 oxidation could lower interfacial shear strength and mitigate ploughing effects. In
7 summary, this analysis elucidates the F_x/F_z increase at 800°C by examining adhesion
8 and ploughing contributions influenced by thermal softening and oxide formation.

9 TEM analysis and molecular dynamics simulations were conducted to further
10 investigate this transformation to examine the subsurface phase transformation
11 mechanisms, as discussed in the following sections.

12

13 **3.2 Phase transformation and defect evolution in subsurface**

14 To further investigate the subsurface structural evolution of monocrystalline
15 silicon under different scratching temperatures, TEM was employed to analyze FIB
16 prepared cross-sectional samples taken 50 μm away from the scratch endpoint in a
17 direction perpendicular to the scratch path. The results provide insights into the phase
18 transformation, defect formation and structural modifications induced by the nanoscale
19 machining process.

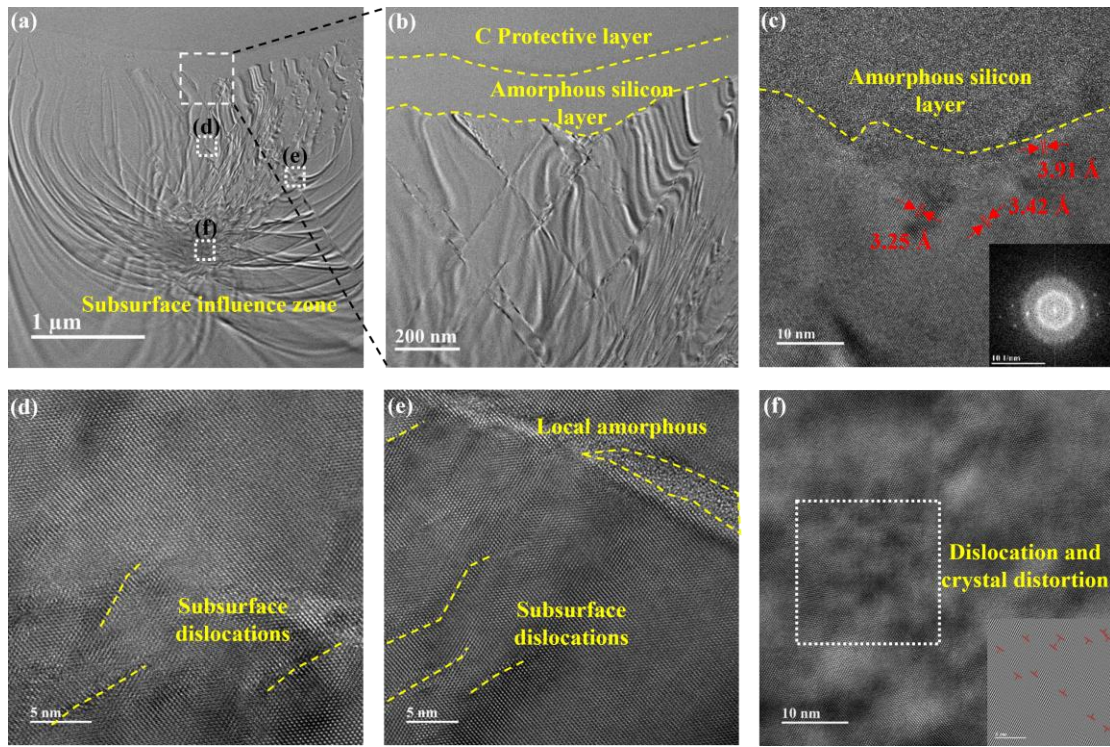


Figure 7. TEM analysis of the subsurface microstructure of monocrystalline silicon after scratching at RT. (a) Low-magnification TEM image showing the subsurface influence zone. (b) High-magnification TEM image of region (b) in (a), revealing a 200 nm-thick amorphous silicon (a-Si) layer formed due to high stress. (c) HRTEM image of the interface between amorphous silicon and monocrystalline silicon at the indentation tip in (b), with an inset showing the corresponding SAED pattern. (d) HRTEM image of region (d) in (a), displaying subsurface dislocations beneath the amorphous layer. (e) HRTEM image of region (e) in (a), illustrating localized amorphization induced by stress concentration at crack tips. (f) HRTEM image of region (f) in (a), highlighting high-density dislocations and lattice distortions in the subsurface region.

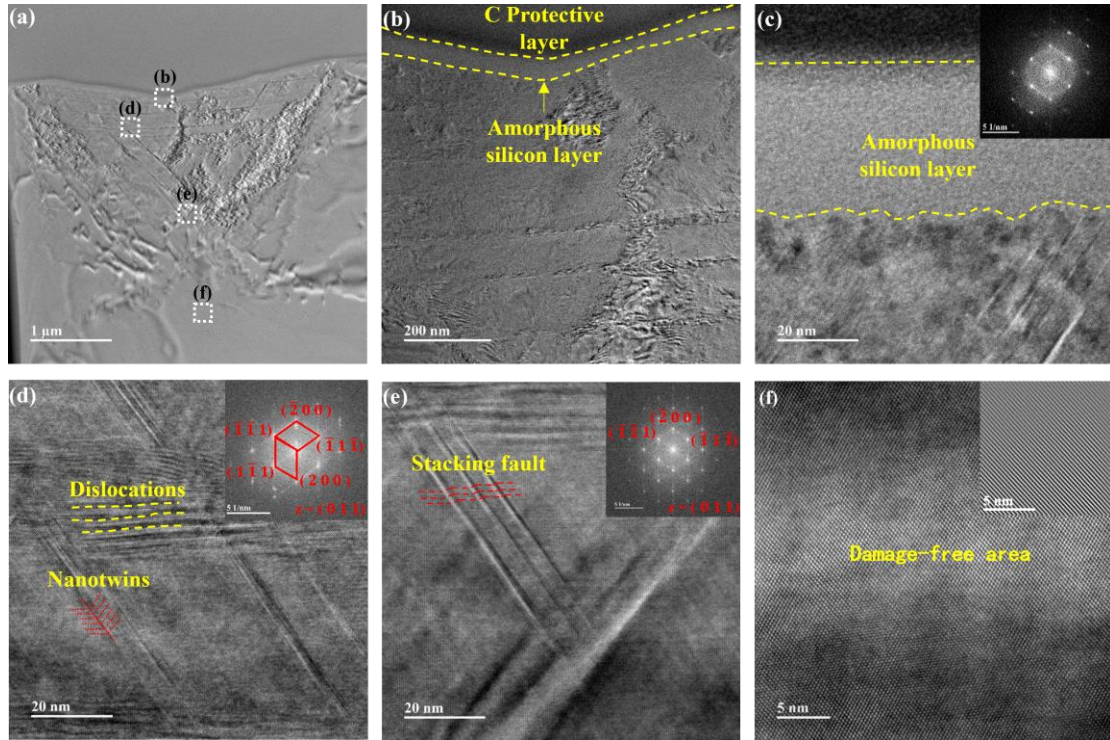
At RT (Fig. 7), the subsurface exhibits a 200 nm-thick amorphous silicon (a-Si)

1 layer formed under scratching-induced stress (Fig. 7(b-c)). High-density dislocations
2 and lattice distortions appear beneath this layer, with localized amorphization at crack
3 tips where stress concentration facilitates phase transformation (Fig. 7(e)). The
4 subsurface influence zone extends beyond 3.8 μm , containing substantial
5 crystallographic defects including dislocations and stacking faults (Fig. 7(d, f)). These
6 observations indicate that RT deformation combines brittle fracture, dislocation
7 generation and stress-induced amorphization, resulting in extensive subsurface damage.

8 At 600°C, significant changes in subsurface microstructure are observed, as shown
9 in Fig. 8. The low-magnification image (Fig. 8(a)) reveals that the extent of the
10 subsurface influence zone is reduced to approximately 2.8 μm compared to 3.8 μm at
11 RT, indicating a more localized deformation effect. Most notably, the thickness of the
12 amorphous silicon layer is substantially reduced to ~35 nm (Fig. 8(b), (c)), suggesting
13 that elevated temperatures suppress the amorphization process. This observation aligns
14 with previous in-situ heating studies[29], which demonstrated that mechanically
15 induced amorphous silicon undergoes epitaxial recrystallization at 600°C, following
16 the original monocrystalline lattice structure. Consequently, the high thermal energy
17 effectively decreases the persistence of amorphization during nanoscale machining.
18 The reduced amorphization at 600°C can be attributed to the competition between
19 stress-induced disorder and thermally-activated recrystallization. The thermal energy
20 facilitates atomic rearrangement back to crystalline configurations during the
21 deformation process itself, unlike at RT where atoms remain trapped in disordered

1 configurations.

2 The 600°C subsurface exhibits characteristic plastic deformation features,
3 including dislocations, stacking faults and nanotwins (Fig. 8(d-e)), indicating a shift
4 toward dislocation-mediated plasticity. The notable absence of microcracks confirms
5 the brittle-to-plastic transition. Fig. 8(f) shows pristine silicon lattice beyond the
6 deformation zone, clearly delineating the influence boundary. This combination of
7 reduced amorphous layer thickness, microcrack absence and well-defined deformation
8 boundary suggests 600°C represents an optimal condition for high-quality machining
9 with minimal subsurface damage. Thermal activation at this temperature promotes
10 defect mobility and recovery processes, mitigating permanent damage while enabling
11 controlled plastic deformation.



13 Figure 8. TEM analysis of the subsurface microstructure of monocrystalline silicon

1 after scratching at 600°C. (a) Low-magnification TEM image showing a significant
2 reduction in defect density compared to RT. (b) High-magnification TEM image of
3 region (b) in (a), revealing a much thinner amorphous silicon (a-Si) layer (~35 nm).
4 (c) HRTEM image of the amorphous silicon layer interface in (b), with an inset
5 showing the corresponding SAED pattern. (d) HRTEM image of region (d) in (a),
6 displaying dislocations and nanotwins. (e) HRTEM image of region (e) in (a),
7 showing stacking faults. (f) HRTEM image of region (f) in (a), a defect-free area.

8

9 As the temperature further increases to 800°C, substantial structural modifications
10 occur, as shown in Fig. 9 and Fig. 10. Energy-dispersive X-ray spectroscopy (EDX)
11 and electron energy loss spectroscopy (EELS) analyses reveal that the surface layer
12 undergoes oxidation, forming an ~20 nm-thick amorphous SiO₂ layer (Fig. 9(b), (c)).
13 The EELS spectrum of silicon exhibits a distinct chemical shift: at depths less than 15
14 nm, the Si⁴⁺ oxidation state is dominant, with the Si L-edge onset at 103–105 eV,
15 characteristic of SiO₂. At 20 nm, the peak gradually shifts from 103 eV to ~100 eV,
16 indicating a transition region where silicon and silicon dioxide coexist. Beyond 20 nm,
17 the spectrum shows a single peak at 99 eV, corresponding to elemental silicon. These
18 findings confirm that the amorphous surface layer consists of SiO₂, while the
19 underlying region retains its silicon structure.

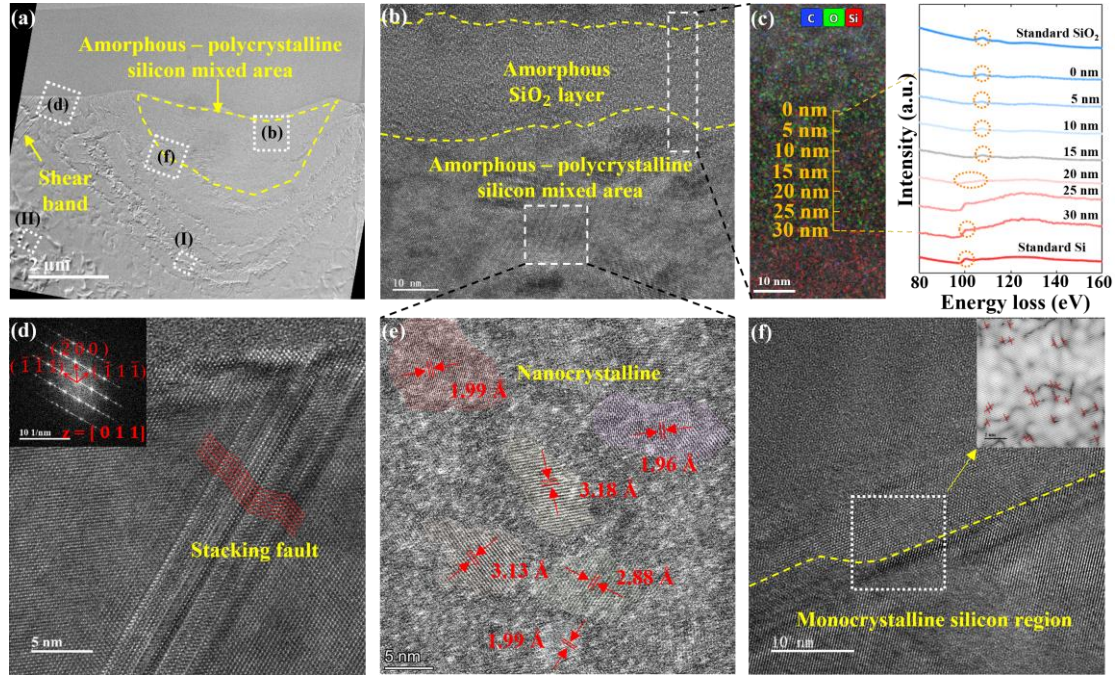
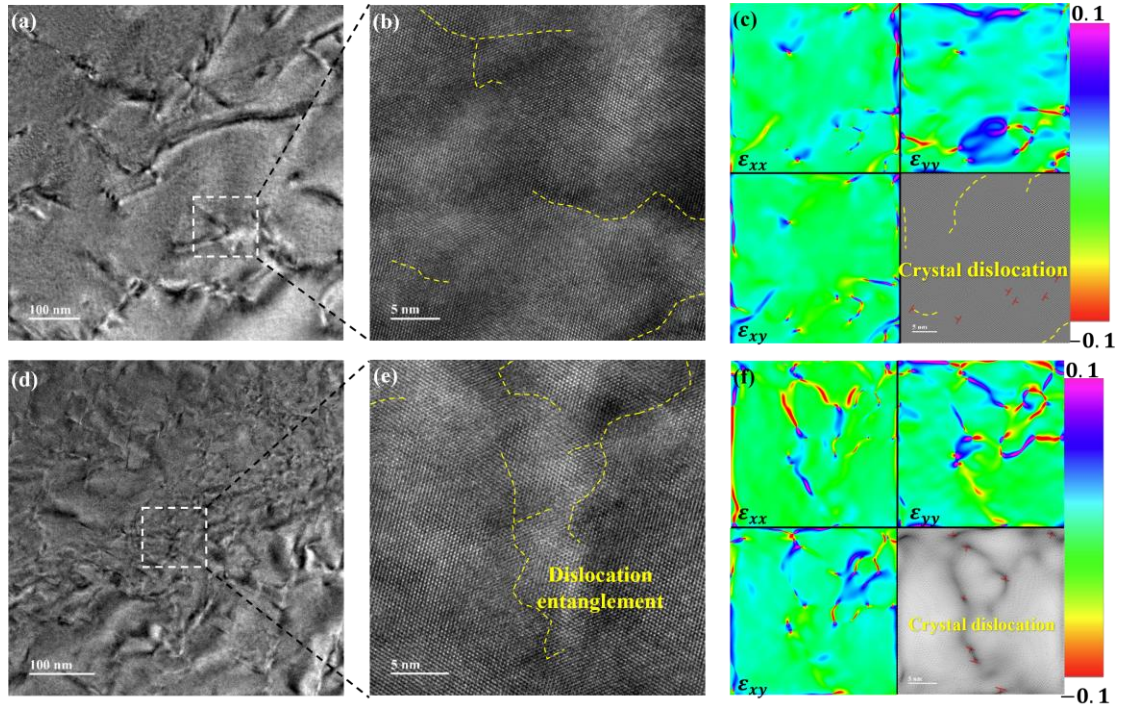


Figure 9. TEM and spectroscopic analysis of the subsurface microstructure of monocrystalline silicon after scratching at 800°C. (a) Low-magnification TEM image showing the formation of an amorphous-polycrystalline silicon mixed regio. (b) High-magnification TEM image of region (b) in (a), revealing the presence of a ~20 nm-thick amorphous SiO₂ layer and the underlying amorphous-polycrystalline silicon region. (c) EDX and EELS analysis confirming the oxidation of the surface layer, with a transition from SiO₂ to elemental silicon at ~20 nm depth. (d) HRTEM image of region (d) in (a), showing stacking faults within the subsurface region. (e) HRTEM image of region (e) in (a), highlighting nanocrystalline silicon formation. (f) HRTEM image of region (f) in (a), showing a transition from the polycrystalline-amorphous region to the underlying monocrystalline silicon, with a high density of dislocations and lattice distortions.

Beneath the SiO₂ layer lies a mixed nanocrystalline-amorphous silicon region (Fig. 9(b, e, f)), attributed to high-temperature softening that facilitates deeper stress-induced amorphization. This mixture forms through rapid nucleation and growth of amorphous silicon coupled with stress-induced distortions. Structural analysis reveals subsurface shear bands (Fig. 9(d)) correlating with extensive shear deformation observed in SEM images (Fig. 5). At the polycrystalline-amorphous/monocrystalline interface, high-density dislocations, and crystal distortions appear (Fig. 9(f)). This suggests that during high-temperature deformation, amorphous silicon initiates recrystallization along original lattice orientations, but stress-induced distortions prevent perfect recovery, resulting in nanocrystalline structure formation rather than complete monocrystalline restoration. The extensive shear bands and nanocrystalline-amorphous mixed region observed beneath the 800°C scratch correlate directly with the pronounced pile-up and plastic flow features observed in the surface morphology analysis (Fig. 5). This subsurface microstructural evidence confirms that the elevated F_x/F_z ratio at 800°C results from the increased resistance to deformation due to extensive material displacement.



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Figure 10. Detailed defect analysis of the subsurface region at 800°C. (a) and (d) TEM images of regions I and II in Fig. 8(a), showing a high density of dislocations. (b) and (e) HRTEM images of the boxed regions in (a) and (d). (c) and (f) Strain distribution maps obtained via geometric phase analysis (GPA), confirming severe lattice distortions and dislocation networks. The absence of defect-free regions indicates that at 800°C, subsurface damage extends beyond the analyzed depth, making it the most severe among the studied temperatures.

Further analysis of the defect structures in Fig. 10 provides additional insights into the subsurface damage at 800°C. The regions marked as I and II in Fig. 9(a) are examined in detail in Fig. 10(a) and (d), respectively, revealing a high density of dislocations. The strain distribution maps obtained via geometric phase analysis

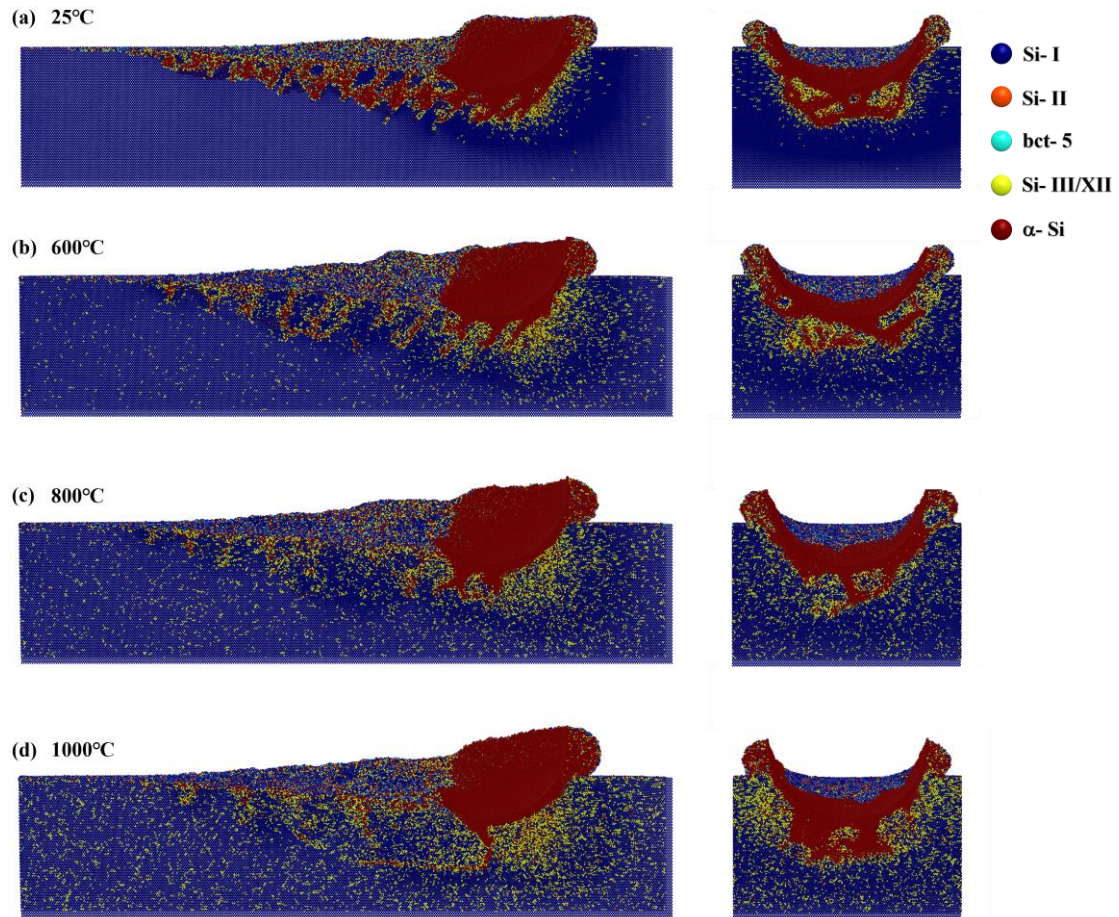
(GPA)[54] (Fig. 10(c) and (f)) confirm that these areas exhibit strain patterns consistent with dislocation networks, indicating severe lattice distortions. The absence of a defect-free region in the TEM samples suggests that at 800°C, subsurface damage extends beyond the analyzed depth, making it the most severe among the three temperatures studied. This extended subsurface alteration at 800°C represents a significant limitation for high-precision silicon processing applications requiring minimal subsurface structural modification.

In summary, subsurface structural evolution during nanoscale machining shows clear temperature-dependent transitions. At RT, brittle fracture and stress-induced amorphization dominate, producing extensive subsurface damage. At 600°C, thermal activation promotes plastic deformation and defect recovery, significantly reducing damage and improving machining quality. At 800°C, excessive thermal softening leads to deep amorphization, oxidation, and polycrystalline formation with severe lattice distortions, creating the least favorable condition for damage-free machining. Molecular dynamics simulations, discussed in subsequent sections, further elucidate these transformation mechanisms.

3.3 Atomistic analysis of phase transformation and force evolution in high-temperature scratching

Molecular dynamics simulations were conducted to investigate atomic-scale phase transformations and subsurface damage evolution at different temperatures. These

1 simulations reveal formation mechanisms of amorphous silicon, dislocation dynamics,
 2 and mechanical responses during nanoscale scratching, complementing experimental
 3 observations to validate deformation mechanisms.



4
 5 Figure 11. Molecular dynamics simulation results showing the phase distribution of
 6 monocrystalline silicon during nanoscale machining at different temperatures. (a) At
 7 25°C, a significant amount of amorphous silicon (red) is generated, penetrating deep
 8 into the subsurface region. (b) At 600°C, the amorphous fraction decreases, and its
 9 penetration depth is reduced. (c) At 800°C, amorphization behavior transitions, with
 10 disordered regions forming both behind and beneath the indenter. (d) At 1000°C, the
 11 amorphous phase predominantly forms toward the inclined front side of the indenter,

1 with a continuous amorphous layer extending deeper into the subsurface.

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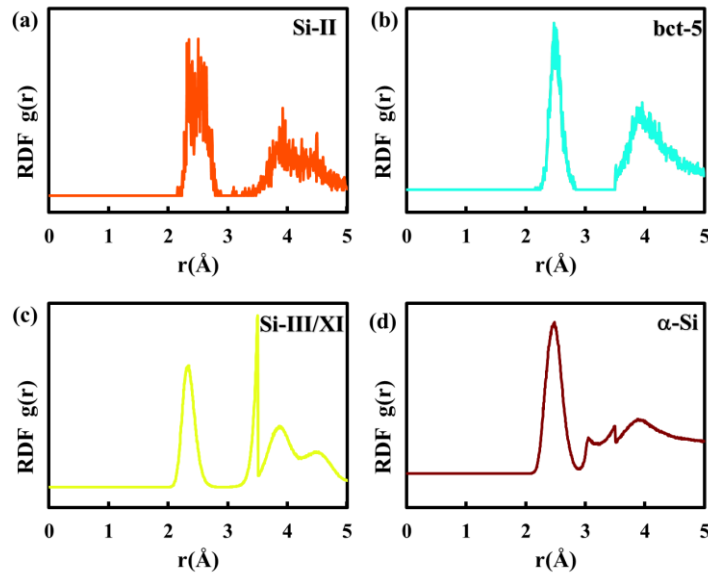
3 As shown in Fig. 11, the atomic-scale phase distribution at different temperatures
4 (25°C, 600°C, 800°C and 1000°C) was analyzed based on atomic coordination numbers
5 and radial distribution functions (RDF) [55]. The identification of silicon phases during
6 nanoscratching relies on the fundamental principle that each crystalline structure
7 exhibits unique atomic arrangements. In our analysis, we employed coordination
8 number (CN) calculations with a cutoff radius of 2.8 Å, combined with RDF profiles
9 to capture these structural signatures.

10 The pristine diamond cubic silicon (Si-I) serves as our reference, characterized by
11 its perfect tetrahedral coordination where each atom bonds with exactly four neighbors
12 at 2.35 Å. This creates a sharp RDF peak followed by a distinctive second peak at 3.84
13 Å representing twelve second-nearest neighbors, a pattern that persists throughout the
14 crystal indicating long-range order. During scratching, this ordered structure transforms
15 dramatically.

16 Fig. 12 presents the quantitative RDF profiles calculated from our MD simulations
17 for the major silicon phases observed during thermal nanoscratching. The high-pressure
18 Si-II phase (Fig. 12(a)) presents a distinctly different signature with its β -tin structure
19 manifesting as six-fold coordination. The RDF reveals a split first-neighbor shell—four
20 atoms at 2.42 Å and two at 2.57 Å—unmistakably identifying this metallic phase
21 formed under the indenter's compressive field. The bct-5 structure (Fig. 12(b)), another

1 high-pressure phase, shows its characteristic RDF pattern with one atom at 2.31 Å and
 2 four atoms at 2.44 Å. The metastable Si-III and Si-XII phases (Fig. 12(c)) maintain four-
 3 fold coordination like Si-I but distinguish themselves through a unique non-bonded
 4 fifth neighbor appearing between 3.2-3.4 Å, absent in the diamond cubic structure.
 5 Amorphous silicon (Fig. 12(d)) emerges as a primary product especially at lower
 6 temperatures, identifiable by its broadened RDF profile where the first peak shifts to
 7 approximately 2.40 Å and subsequent peaks progressively fade.

8 The results indicate that at RT, a significant amount of amorphous silicon is
 9 generated, penetrating deep into the subsurface region. As the temperature increases,
 10 the fraction of amorphous silicon decreases, and its penetration depth first reduces from
 11 RT to 800°C but then increases significantly at 1000°C. This trend aligns with
 12 experimental observations, where subsurface damage was minimized at 600–800°C but
 13 became more pronounced at higher temperatures.



1 Fig. 12. Radial distribution functions (RDF) from MD simulations for different
2 silicon phases: (a) Si-II (β -tin structure), (b) bct-5 structure, (c) Si-III/XII phases, and
3 (d) amorphous silicon. These distinctive RDF patterns serve as structural fingerprints
4 for phase identification during thermal nanoscratching.

5 At lower temperatures (25-600°C), the amorphous phase primarily forms behind
6 the indenter, growing toward the inclined rear side opposite to the scratching direction,
7 indicating dominant roles of shear and tensile stresses in amorphization. At 800°C, a
8 transition occurs—amorphous silicon is no longer confined to the rear direction but
9 forms more complex disordered regions both behind and beneath the indenter,
10 reflecting competition between mechanical amorphization and thermal recrystallization
11 processes.

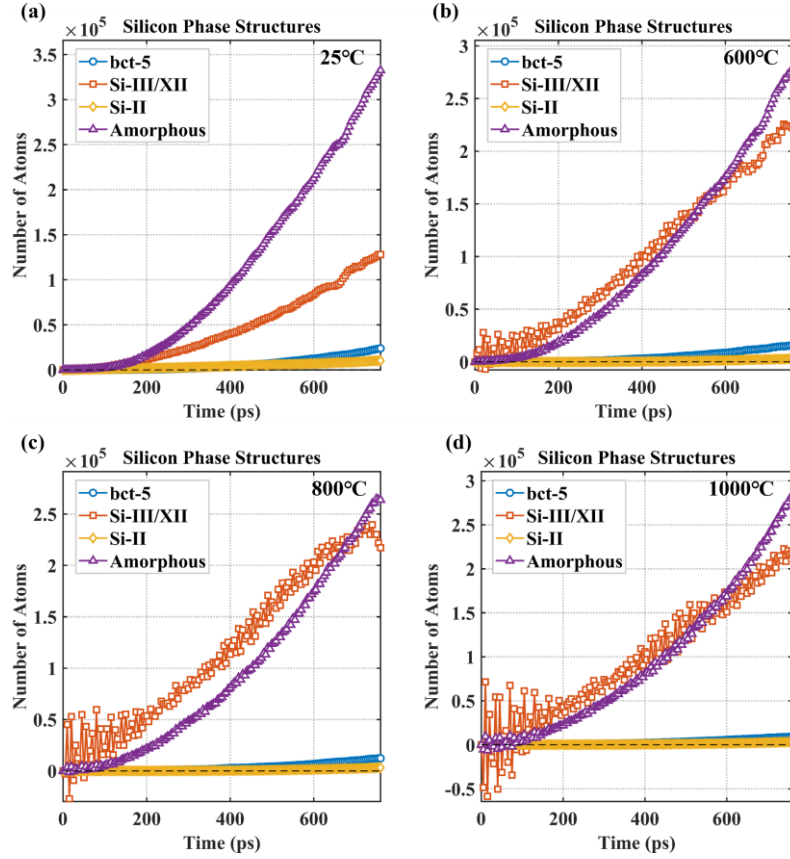


Figure 13. Molecular dynamics simulation results of phase evolution during nanoscale machining of monocrystalline silicon at different temperatures. (a)-(d) Evolution of different silicon phase structures (bct-5, Si-III/XII, Si-II and amorphous) over time at 25°C, 600°C, 800°C and 1000°C.

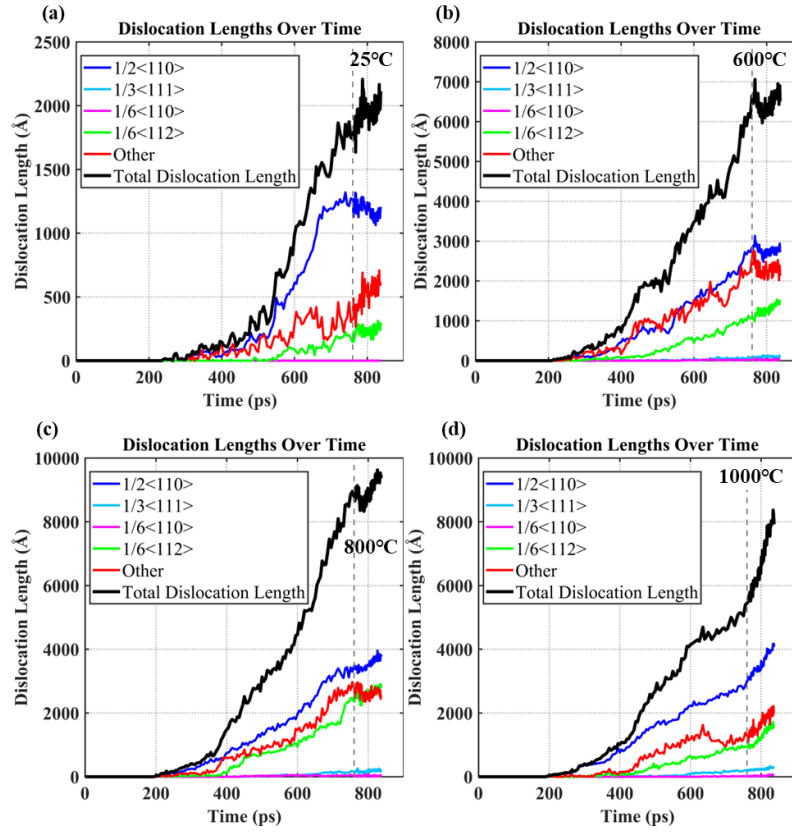
At 1000°C, amorphization undergoes another shift—forming predominantly toward the indenter's front side with a continuous layer extending deeper into the subsurface. This indicates hydrostatic pressure at the leading edge becomes the primary amorphization mechanism, replacing the shear/tensile-driven processes observed at lower temperatures. Pre-existing amorphous regions provide nucleation sites for further

1 amorphization under higher loads, leading to deeply penetrating amorphous-
2 polycrystalline mixed regions similar to experimental observations. The phase
3 distribution also transitions from symmetric (25°C) to asymmetric and localized (600-
4 1000°C), reflecting the changing dominant amorphization mechanisms. The statistical
5 analysis of phase and dislocation evolution, presented in Fig. 13, further supports these
6 findings. It should be noted that for phase structure analysis, we assumed that all atoms
7 were in the perfect Si-I phase at the beginning of machining, representing an
8 undamaged initial state. Therefore, all phase counts were zeroed at the starting point,
9 which can occasionally lead to fluctuations resulting in negative values at certain
10 timepoints (such as for Si-III/XII at 1000°C) due to thermal effects. The number of
11 amorphous atoms first decreases with increasing temperature, dropping from 3.3×10^5
12 at RT to 2.7×10^5 at 600°C and further to 2.6×10^5 at 800°C. However, at 1000°C, the
13 amorphous atom counts increases again to 2.8×10^5 , confirming the re-intensification
14 of amorphization at extreme temperatures. Additionally, the fraction of bct-5 phase
15 decreases with temperature, while high-temperature conditions favor the
16 transformation into Si-III/XII phases, indicating a shift in phase transformation
17 pathways. This trend is consistent with previous studies[28,56], which have also
18 reported that elevated temperatures promote the formation of high-pressure silicon
19 phases such as Si-III and Si-XII, while suppressing the formation of bct-5.

20 These results demonstrate clear temperature-dependent transitions in phase
21 transformation mechanisms. At lower temperatures ($\leq 600^\circ\text{C}$), hydrostatic pressure and

1 shear stress behind the indenter drive amorphization primarily toward the rear side
2 opposite the scratching direction. With rising temperature, silicon atoms gain sufficient
3 energy for recrystallization, significantly reducing amorphous phase formation through
4 post-deformation recovery into more ordered crystalline states.

5 Dislocations evolution analysis (Fig. 14) shows total dislocation length initially
6 increases with temperature, indicating enhanced plastic deformation, before
7 unexpectedly decreasing at 1000°C. At this extreme temperature, dislocation length
8 increases sharply after indenter lifting (marked by gray dashed lines), attributed to
9 recrystallization and mechanical loading producing randomly oriented nanocrystals
10 whose grain boundaries hinder dislocation propagation. Importantly, this reduced
11 dislocation length does not indicate decreased subsurface damage—rather, the
12 formation of nanocrystalline regions coupled with increased amorphization confirms
13 that damage is significantly exacerbated at 1000°C, making it least favorable for
14 damage-free machining.



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Figure 14. Molecular dynamics simulation results of dislocation evolution during nanoscale machining of monocrystalline silicon at different temperatures. (a)-(d) Evolution of total dislocation length and individual dislocation types ($1/2\langle 110 \rangle$, $1/3\langle 111 \rangle$, $1/6\langle 110 \rangle$, $1/6\langle 112 \rangle$ and other) over time at 25°C, 600°C, 800°C and 1000°C.

Analysis of dislocation types (Fig. 14(a-d)) reveals significant temperature-dependent variations. At 25°C, $1/2\langle 110 \rangle$ perfect dislocations dominate while $1/6\langle 110 \rangle$ and $1/6\langle 112 \rangle$ partial dislocations remain minor. With increasing temperature (600-

1 800°C), total dislocation length increases while the proportion of $1/6\langle 112 \rangle$ partial
2 dislocations gradually rises, though $1/2\langle 110 \rangle$ dislocations remain dominant throughout.

3 As further visualized in Fig. 15, the spatial distribution of these dislocations
4 exhibits distinct temperature-dependent patterns. At room temperature (25°C),
5 dislocations are primarily concentrated in a limited region and predominantly extend
6 toward the oblique rear direction of the indenter. This directional preference is driven
7 by the combined action of shear and tensile stresses in this region, creating favorable
8 conditions for dislocation nucleation and propagation. This observation aligns precisely
9 with our TEM findings (Fig. 8) showing high-density dislocations beneath the 200 nm-
10 thick amorphous silicon layer formed during RT scratching. The localized
11 amorphization at crack tips (Fig. 8(e)) further corroborates the simulation results,
12 demonstrating that at low temperatures, silicon's deformation is characterized by brittle
13 fracture, limited dislocation mobility, and stress-induced amorphization due to high
14 Peierls barriers[26].

15 At 600°C, a notable change occurs with the emergence of more partial dislocations
16 that significantly enhance atomic mobility. The dislocation networks not only increase
17 in density but also spread more extensively into the subsurface volume, forming
18 interconnected structures that facilitate material flow. This computational observation
19 perfectly complements our TEM analysis (Fig. 9(d)-(e)), where characteristic defect
20 structures including dislocations, stacking faults and nanotwins dominate the
21 subsurface region while the amorphous layer thickness is drastically reduced to ~35 nm.

1 The expansion of dislocation networks becomes even more pronounced at 800°C,
2 where dislocation networks develop far greater complexity. Unlike the room
3 temperature case, dislocations are no longer confined to propagating mainly in the
4 oblique rear direction of the indenter but also extend toward the front of the indenter.
5 This bidirectional propagation indicates that defect formation is increasingly influenced
6 by compressive and shear stresses acting simultaneously, demonstrating the thermal
7 activation of multiple slip systems. Our TEM observations (Fig. 10-11) provide striking
8 experimental confirmation of this phenomenon, revealing extensive shear bands (Fig.
9 10(d)) and high-density dislocation networks that extend beyond the analyzed depth.
10 The geometric phase analysis (GPA) strain maps (Fig. 11(c), (f)) quantitatively validate
11 the severe lattice distortions predicted by our simulations, offering microscopic
12 evidence for the macroscopic plastic flow features observed in surface morphology.

13 At 1000°C, we observe an interesting phenomenon in simulations where
14 dislocations near the surface encounter obstacles due to intensive phase transformations,
15 hindering their propagation. However, dislocations in deeper regions extend farther and
16 penetrate deeper into the substrate, forming the most extensive three-dimensional
17 networks across all tested temperatures. While our experiment and TEM analysis did
18 not extend to 1000°C, this simulation finding is consistent with the trend observed at
19 800°C.

20 In our MD simulations, we incorporated a 1nm-thick thermostatic layer to
21 replicate the constant temperature conditions of the actual experiments. Although local

1 temperature increases may occur during the scratching process in both simulations and
2 real experiments, our simulations were specifically designed to investigate the
3 instantaneous material response at designated temperatures, making the effects of local
4 temperature rises beyond the scope of our discussion.

5 This evolution reflects two competing mechanisms: (1) elevated temperatures
6 activate multiple slip systems and promote dislocation nucleation across various
7 crystallographic orientations, and (2) thermal energy enhances dislocation mobility,
8 cross-slip and partial dislocation formation, particularly benefiting $1/6\langle 112 \rangle$ partial
9 dislocations associated with stacking faults and nanotwins.

10 The transition from directionally constrained dislocation propagation at room
11 temperature to complex multi-directional networks at elevated temperatures illustrates
12 how thermal energy reduces Peierls barriers and activation energies for dislocation
13 motion[57]. This thermal facilitation allows dislocations to overcome local energy
14 barriers more easily, navigate through the crystal lattice along multiple slip systems,
15 and interact to form complex substructures that effectively distribute plastic
16 deformation throughout a larger volume. Our TEM observations at 600°C (Fig. 9(d)-
17 (e)) provide direct visual evidence of this facilitated defect mobility, revealing well-
18 defined stacking faults and nanotwins that are characteristic signatures of partial
19 dislocation activity.

20 The increased $1/6\langle 112 \rangle$ partial dislocation activity is particularly significant.
21 These partial dislocations exhibit lower energy barriers for motion compared to perfect

1 dislocations, facilitating atomic rearrangement and enhancing plasticity[58]. The
2 formation of twins with $\{111\}$ habit planes and the splitting of perfect dislocations into
3 Shockley partials (following the scheme $1/2\langle 110 \rangle \rightarrow 1/6\langle 112 \rangle + 1/6\langle 121 \rangle$) becomes
4 energetically favorable at elevated temperatures. This mechanism correlates with our
5 observations of increased $1/6\langle 112 \rangle$ partial dislocation activity and explains how their
6 increased mobility promotes plastic flow rather than brittle fracture, contributing to the
7 observed temperature-dependent mechanical behavior during nanoscratching.

8 The impediment of dislocation propagation near the surface at elevated
9 temperatures (particularly at 1000°C in simulations) further highlights the complex
10 interplay between phase transformations and dislocation dynamics. This phenomenon
11 may relate to the competition between amorphization and crystallization processes we
12 observed experimentally at 800°C, where the formation of a nanocrystalline-amorphous
13 mixed region (Fig. 10(e)) indicates that stress-induced distortions prevent perfect
14 recovery during recrystallization. This experimental finding provides physical context
15 for the computational observation that high-temperature phase transformations can
16 hinder dislocation propagation near the surface.

17 The quantitative dislocation analysis in Fig. 14 provides comprehensive
18 computational evidence for temperature-enhanced plasticity mechanisms that are
19 directly validated by our TEM observations.

20 In conclusion, our atomic-scale simulations provide critical insights that
21 complement our experimental findings, revealing temperature-dependent deformation

1 mechanisms in monocrystalline silicon. The simulations capture key transitions from
2 stress-induced amorphization and limited dislocation activity at 25°C to enhanced
3 dislocation mobility with reduced amorphization at 600°C, followed by complex
4 bidirectional dislocation networks and competing phase transformations at higher
5 temperatures. The quantitative analysis of phase evolution and dislocation dynamics
6 confirms 600°C as the optimal temperature for silicon machining, where thermal energy
7 effectively suppresses brittle fracture while enhancing plasticity without excessive
8 thermal softening or oxidation. Further investigations with larger-scale simulations and
9 experimental validations across varied machining rates are needed to fully understand
10 plastic flow and chip formation mechanisms at elevated temperatures.

11

4. Conclusions

This study systematically investigated the tribological behavior and subsurface phase evolution of monocrystalline silicon during high-temperature nanoscratching (RT~800°C), revealing nanoscale transition mechanisms in material removal and subsurface defect formation under thermo-mechanical coupling. Key findings include:

(1) Material removal mechanisms transition from brittle fracture (RT-400°C) to ductile ploughing (600°C) to excessive plastic flow (800°C). This transition correlates with a stable lateral force-to-normal force ratio ($F_x/F_z \approx 0.3$) from RT to 600°C that sharply increases to 0.78 at 800°C due to thermally-enhanced plasticity and oxide film formation that significantly affect tribological behavior. The fundamental deformation mechanism shifts from phase transformation-mediated plasticity at lower temperatures to dislocation-dominated deformation at elevated temperatures, as evidenced by the extensive dislocation networks and shear bands observed at 800°C.

(2) Subsurface TEM characterization showed temperature-dependent defect evolution. At RT, a crystal defect layer comprising amorphous Si and dislocations extended beyond 3.8 μm beneath the scratch. At 600°C, the defect depth decreased to ~2.8 μm with the amorphous Si layer thickness reducing to merely 1/6 of that observed at RT, due to thermally activated crystal healing and epitaxial recrystallization. At 800°C, a 20 nm-thick SiO_2 surface layer formed, beneath which a mixed nanocrystalline-amorphous Si structure developed with more extensive subsurface

1 damage. These observations identify around 600°C as an optimal temperature range for
2 minimizing subsurface defects in silicon nanomachining.

3 (3) Molecular dynamics simulations demonstrated that the activity of partial
4 dislocations (primarily $1/6\langle 112 \rangle$) increased with temperature, correlating with the
5 experimental observation of nanotwins and stacking faults at 600~800°C. In addition,
6 a transition in the amorphization mechanism was identified: from RT to 600°C,
7 amorphization occurs primarily through combined shear stress and tensile stress behind
8 the indenter during nanoscratching, while at 800°C and above, hydrostatic stress at the
9 indenter front becomes the dominant cause of amorphization. Elevated temperatures
10 also promote Si-III/XII phase formation while suppressing the bct-5 phase formation
11 during nanoscratching.

12 This study identifies around 600°C as a favorable processing temperature range
13 for nanoscale machining of monocrystalline silicon, where enhanced plasticity
14 combines with minimal subsurface damage. The research provides insights for
15 optimizing thermal conditions in the precision manufacturing of silicon-based devices.

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2

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